

Pitch and Sell — Exhaustive Feature & Architecture Documentation

This document serves as an exhaustive breakdown of absolutely every feature, screen, component, and architectural system built into the **Pitch and Sell** platform (Backend & Mobile App).

1. Authentication & Onboarding

Welcome Screen (UI/UX)

- **Dynamic Tabs:** Switch smoothly between 'Sign In' and 'Sign Up' forms.
- **Email Sign-In:** Email and password fields with a visibility toggle (eye icon).
- **Email Sign-Up:** Comprehensive form capturing First Name, Last Name, Email, Phone Number, Password, and Profile Picture (via device gallery).
- **Role Selection:** Users must define their intent via a dropdown: `Buyer` , `Seller` , or `Both` .
- **Social Connect:** One-tap OAuth buttons for Google and Facebook.
- **Validation:** Strict frontend validation preventing empty fields and unchecked "Terms & Conditions" boxes.

Backend Auth Architecture

- **Password Encryption:** Uses `bcryptjs` to securely hash passwords before storing them in Supabase.
 - **JWT Sessions:** Returns a secure JSON Web Token upon login.
 - **Google OAuth Verification:** Backend utilizes `google-auth-library` to securely verify the `idToken` generated by the Flutter app.
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2. Main Navigation & Core UI

- **Persistent Bottom Navigation Bar:** Gives access to Home (Feed), Explore (Search), Live/Upload, Inbox (Chats), and Profile.
- **Global Role Switcher:** A prominent toggle button in the top navigation allowing dual-role accounts to instantly switch between the "Buyer" experience and the "Seller Dashboard" without logging out.

- **Shimmer Loaders:** Premium skeleton loading animations are displayed while fetching data, completely replacing basic circular spinners.
 - **Deep Linking:** Configured with `app_links` to intercept URLs (e.g., `https://pitch-and-sell-backend.onrender.com/product/123`) and open specific videos directly in the app.
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3. The Video Commerce Feed (TikTok-Style)

Video Player

- **Vertical Paging:** `PageView.builder` enables smooth, infinite vertical swiping.
- **Pre-Caching Engine:** Utilizes `flutter_cache_manager` to download and cache videos to the local device storage, eliminating buffering on subsequent views.
- **Gestures:** Tap to Play/Pause; Double-Tap to Like (triggers a floating, fading red heart animation).
- **Lifecycle Management:** Videos automatically pause when swiped away or when the app is minimized.

Interactive Overlays

- **Sticky Header:** Contains Search, Cart, unread Notification Bell, and the Buy/Sell mode toggle.
- **Categories Bar:** Horizontal scrollable pills to filter the feed (e.g., All, Fashion, Tech, Food, Handmade).
- **Action Column:** Floating buttons on the right side for Profile, Likes (with counter), Comments, and Share.
- **Share:** Copies a dynamic link to the clipboard to share the product externally.

Bottom Sheet Commerce Integration

- **Product Details Sheet:** Pulls up over the video displaying Name, Price, Discount/Old Price banner.
 - **Quantity Selector:** Interactive (+/-) counter bounded between 1 and 24.
 - **Color/Variant Selector:** Selectable pills for product variations (e.g., Black, Silver, Orange).
 - **Seller Verification Badge:** Shows the seller's mini-profile and a "Verified Business Seller" green checkmark.
 - **Action Buttons:** "Add to Cart" (shows a success toast) and "Buy Now" (routes directly to checkout).
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4. E-Commerce Engine (Cart & Checkout)

- **Cart Screen:** Lists all accumulated products with individual quantity adjusters, subtotal calculations, and swipe-to-delete.
 - **Checkout Flow:**
 - Displays Order Summary.
 - Form for Shipping Address.
 - Payment Method selection (Cash on Delivery, Credit/Debit Card, Mobile Wallet).
 - **Order Management Screen:** Buyers can view "Active" and "Past" orders, track fulfillment status, and cancel pending orders.
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5. Seller Dashboard & Analytics

Financial Ledger

- **Available Balance:** Large central display of withdrawable funds.
- **Analytics Grid:** 4 data cards tracking Gross Sales, Platform Fees, Net Earnings, and Total Paid Out.
- **Audit Logs:** A chronological list of every ledger event (e.g., `SALE_EARNING` , `WITHDRAWAL`), color-coded green for credits and red for debits.
- **Payout System:** Sellers can trigger a withdrawal request via a bottom sheet, selecting from Bank Account, EasyPaisa, or JazzCash, and providing their account details (Enforces a minimum PKR 500 threshold).

Content Management

- **Upload Video Flow:** A comprehensive form to publish a new pitch. Fields include Product Name, Price, Description, Category, Stock Qty, and a Video Picker. Includes a toggle to allow/disallow viewers from downloading the video.

AI Pitch Generator 🤖

- **Inputs:** Sellers input their Product Name, Key Selling Point (e.g., "Active Noise Cancellation"), Tone (Exciting vs. Professional), and Language (English or Roman Urdu).
- **Output:** Queries the backend AI route to generate a high-converting, 15-second video script formatted beautifully in monospace font.
- **Recording Tips:** Returns a bulleted list of camera and lighting tips based on the product type.
- **Clipboard Integration:** One-tap button to copy the script to the clipboard for teleprompter apps.

Boost Promotion Engine (Monetization)

Sellers can pay to boost their video's algorithm ranking:

- **Standard Tier:** 2x visibility multiplier for 24h (Rs 500).

- **Premium Tier:** 5x visibility multiplier for 72h (Rs 1,500).
 - **Elite Tier:** 10x visibility multiplier for 7 days (Rs 5,000).
 - **Active Promos Tracker:** A card on the dashboard showing the active boost's remaining time, total impressions, clicks, and orders generated.
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6. Social & Real-Time Features

- **Real-Time Chat:** Powered by `Socket.io`. Users can join specific chat rooms, send messages instantly, and receive messages without pulling to refresh.
 - **Comments Section:** A bottom sheet allowing users to read and post comments on specific videos.
 - **Notifications Screen:** Central hub for order updates, likes, and messages.
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7. Backend Infrastructure & Security

Technology Stack

- **Server:** Node.js with Express.js, written fully in strict TypeScript.
- **Database:** PostgreSQL hosted on Supabase (accessed via IPv4 Session Pooler).
- **ORM:** Prisma Client for strict database typing and migrations.
- **Media Handling:** `multer` configured for local/static file uploads (videos/images).

Security Architecture

- **Rate Limiting:** `express-rate-limit` caps every IP address to 100 requests per 15 minutes to prevent DDoS and Brute Force attacks.
- **Helmet:** Automatically injects 10+ HTTP security headers to prevent XSS, Clickjacking, and sniffing.
- **Global Error Handling:** All errors are caught by `error.middleware.ts`, preventing the server from crashing and ensuring the mobile app receives predictable `{ success: false, error: "..."}` JSON responses.

Server Routing & APIs

- `/api/auth`: `POST /signup`, `POST /signin`, `POST /google`, `GET /me`, `PUT /profile`.
- `/api/business`: Endpoints for creating and updating the business/seller KYC profile.
- `/api/products`: `POST /` (Upload with Video), `GET /feed` (Algorithm logic for categories and search), `POST /video/:id/like`, `GET/POST /video/:id/comments`, `POST /:id/promote` (Boost).

- **/api/orders** : POST / (Create), GET / (Fetch by mode), PATCH /:id/status , POST /:id/cancel .
 - **/api/orders/ledger** : Returns the seller's financial summary and audit trail.
 - **/api/orders/payout** : Initiates a withdrawal request.
 - **/api/ai/generate-pitch** : Handles LLM prompt construction for the Pitch Generator.
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8. Enterprise Readiness (Firebase)

The Flutter app (`main.dart`) is pre-coded and prepared for integration with the Firebase suite:

- **Firebase Crashlytics**: Will automatically catch fatal Dart errors and native crashes, reporting them to the cloud.
- **Firebase Analytics**: Prepared to track screen views and video retention times.
- **Firebase Cloud Messaging (FCM)**: Ready to accept permissions for native OS push notifications so sellers are alerted of new orders instantly.